

Chapter 16, Problem 2P

Problem

The differential equation that describes the voltage in an RLC network is

$$3\frac{d^2i}{dt^2} + 15\frac{di}{dt} + 12i = 0$$

Given that $v(0) = 0$, $dv(0)/dt = 6$ mA/s, obtain $i(t)$.

Step-by-step solution

Step 1 of 5

The differential equation for the voltage in the RLC circuit is,

$$3\frac{d^2i}{dt^2} + 15\frac{di}{dt} + 12i = 0$$

But for the voltage in the RLC circuit, the differential equation should be in terms of v .

Hence consider that the differential equation for the voltage in the RLC circuit is,

$$3\frac{d^2v}{dt^2} + 15\frac{dv}{dt} + 12v = 0 \dots\dots (1)$$

The initial voltage conditions are, $v(0) = 0$ and $\frac{dv(0)}{dt} = 6$.

Step 2 of 5

Apply Laplace Transform to the equation 1.

$$3\left(s^2V(s) - sv(0) - \frac{dv(0)}{dt}\right) + 15(sV(s) - v(0)) + 12V(s) = 0$$

Substitute 0 for $v(0)$ and 6 for $\frac{dv(0)}{dt}$.

$$3(s^2V(s) - 6) + 15sV(s) + 12V(s) = 0$$

$$(3s^2 + 15s + 12)V(s) - 18 = 0$$

$$(s^2 + 5s + 4)V(s) = 6$$

$$\begin{aligned} V(s) &= \frac{6}{s^2 + 5s + 4} \\ &= \frac{6}{(s+1)(s+4)} \end{aligned}$$

Step 3 of 5

Consider partial fraction expansion.

$$V(s) = \frac{A}{(s+1)} + \frac{B}{(s+4)} \dots\dots (2)$$

$$\frac{6}{(s+1)(s+4)} = \frac{A}{(s+1)} + \frac{B}{(s+4)}$$

$$6 = A(s+4) + B(s+1) \dots\dots (3)$$

Substitute -1 for s .

$$6 = A(-1+4) + B(0)$$

$$\begin{aligned} A &= \frac{6}{3} \\ &= 2 \end{aligned}$$

Substitute -4 for s in equation 3.

$$6 = A(0) + B(-4+1)$$

$$\begin{aligned} B &= \frac{-6}{3} \\ &= -2 \end{aligned}$$

Step 4 of 5

Substitute 2 for A and -2 for B in equation 2.

$$V(s) = \frac{2}{s+1} - \frac{2}{s+4}$$

Apply Inverse Laplace Transform.

$$L^{-1}[V(s)] = L^{-1}\left[\frac{2}{s+1}\right] - L^{-1}\left[\frac{2}{s+4}\right]$$

$$\begin{aligned} v(t) &= 2e^{-t}u(t) - 2e^{-4t}u(t) \\ &= 2(e^{-t} - e^{-4t})u(t) \text{ V} \\ &= 2(e^{-t} - e^{-4t}) \text{ V} \quad \text{for } t > 0 \end{aligned}$$

Thus, the $v(t)$ for $t > 0$ is $\boxed{2(e^{-t} - e^{-4t}) \text{ V}}$.

Step 5 of 5

If we consider that the differential equation for the voltage in the RLC circuit is,

$$3 \frac{d^2 i}{dt^2} + 15 \frac{di}{dt} + 12i = 0$$

Follow the process to find $v(t)$ and replace v by i .

Then the $i(t)$ for $t > 0$ is $\boxed{2(e^{-t} - e^{-4t}) \text{ A}}$.